Where the Water Goes

Targeting the on-farm modernisation budget using the WaPOR evaporation–transpiration split · Summer (Nili/Sefi) 2024 · prepared for the budget allocation committee

1. The decision this brief supports

The Ministry of Water Resources and Irrigation allocates a fixed annual on-farm modernisation budget across governorates and must justify that spending. IPAT, the irrigation assessment tool already operating on the ministry platform, shows *where* water productivity is high and low across roughly 1.5 million hectares of the Middle and West Delta. It does not indicate where a unit of spending recovers the most water. This brief supplies that ordering.

2. What was measured

FAO WaPOR v3 Level 2 (100 m) dekadal actual evapotranspiration and interception, evaporation, transpiration and interception, summed over 18 dekads covering 1 May – 31 October 2024. Cropland was separated from desert by a seasonal AETI threshold of 200 mm; open water was excluded by JRC Global Surface Water occurrence, by a beneficial-fraction floor, and by a physical ceiling on cropland evaporation. Observed irrigated area totalled 3.56 million hectares against 3.21 million hectares of reported cultivated area.

Across the study area, 4,844 Mm³ of water evaporated during the season without growing a crop. That water was delivered and paid for. A share of it is recoverable through low-cost on-farm measures.

3. Recommended allocation of a $12.0M budget

#	GOVERNORATE	MEASURE	FUNDED	HECTARES	MM³/YR	$/M³
1	Aswan	Laser land levelling	$1.19M	79,350	29.5	$0.040
2	Qena	Laser land levelling	$2.74M	182,502	67.8	$0.040
3	Sohag	Laser land levelling	$2.17M	144,522	50.8	$0.043
4	Assiut	Laser land levelling	$2.23M	148,668	48.3	$0.046
5	Beni Suef	Laser land levelling	$1.97M	131,054	41.7	$0.047
6	Qalyubia	Laser land levelling	$1.12M	74,955	22.9	$0.049
7	Minya	Laser land levelling	$0.58M	38,949	11.8	$0.049

Targeted allocation recovers 273.0 Mm³ at $0.044 per cubic metre. The same budget distributed in proportion to irrigated area recovers 218.1 Mm³ — so ordering the spending yields **25% more water for the same money**, and the same figure on unit cost.

4. Equity is priced, not assumed away

Spending strictly cheapest-water-first concentrates the budget in a handful of governorates. An equity floor reserves a share of the budget to be distributed across every governorate by irrigated area before the remainder is targeted. The cost of that choice is quantified rather than argued.

EQUITY FLOOR	GOVERNORATES FUNDED	WATER RECOVERED, MM^3	COST VS. PURE TARGETING

5. How much of this requires the satellite

The allocation was re-run with the satellite information progressively removed. With per-district evaporation, targeting gains —%. Knowing only a per-irrigation-system average it gains —%. With evaporation held spatially uniform it gains —%. The advantage is therefore entirely spatial, but most of it resides in the system-level pattern rather than in district-level detail.

6. Limitations

Field scale is not basin scale. In a closed system such as the Nile Delta, a share of the water an intervention frees was already returning to drains and aquifers and being reused downstream. Suppressing soil evaporation can also convert it to transpiration and yield rather than to recoverable water. Recoverable evaporation is an upper bound, and these figures rank locations against one another; they do not constitute a water account.

Retrieval uncertainty is concentrated where the signal is. The separation of evaporation from transpiration is the least constrained element of the underlying energy-balance model, particularly over standing water and partially covered soil. Chukalla et al. (2022) excluded the beneficial fraction from their WaPOR irrigation performance framework for this reason. The ordering presented here should be treated as a prioritisation hypothesis to be confirmed by field measurement, not as a settled result.

Cost coefficients are literature-order estimates. Intervention costs, service lives and evaporation-reduction fractions are not measurements. The ranking of locations is driven by observed evaporation and is comparatively stable; the magnitude of the prize moves with every coefficient and should be quoted as an order of magnitude.

The comparator is a neutral prior. Area-proportional allocation represents what one would do knowing only where the cropland lies. It is not a description of current ministry practice, for which no citation is claimed.

The decision unit is administrative. Governorates are not how this budget is delivered; irrigation directorates and command areas are. The method is boundary-agnostic and runs unchanged on any polygon set, including those used by IPAT.

7. Monitoring

Because the indicator that justifies the spending is also the indicator that measures it, the following season's WaPOR imagery observes funded and unfunded governorates under the same instrument and at the same cost. Funded and unfunded areas form a natural comparison, permitting a difference-in-differences estimate of effect with a stated confidence interval — including the case in which the effect is too small to detect.

FAO WaPOR v3 Level 2 · 18 dekads · 1 May – 31 Oct 2024 · retrieved 19 August 2026 · method after Chukalla et al. (2022) · generated from the interactive tool, all coefficients editable and listed in the accompanying documentation.